

Stress Level Classifier Using Text Data

Phase 1 Report — Dataset Formation and Initial Analysis

CSCI417 — Machine Intelligence

Medical Track — Stress Level Classifier

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Abstract

This report presents the work completed in Phase 1 of the Stress Level Classifier project, developed as part of CSCI417 — Machine Intelligence. The primary objective of Phase 1 was to construct a labeled text dataset suitable for training machine learning models to classify written text into one of three stress levels: Low, Medium, and High.

The dataset was assembled from two distinct sources. The first source is the MentalDistress dataset, a publicly available collection of text entries associated with mental health labels, from which three relevant labels were selected and mapped to the target stress categories. The second source consists of text data collected from Stack Exchange using its public API, providing real-world examples of stress-related discourse across academic, professional, and social contexts.

Following data cleaning, label mapping, deduplication, and balancing, the final dataset contains 5,523 samples distributed equally across the three classes, with 1,841 samples per class. The dataset is stored in a structured CSV file containing fifteen metadata columns to support traceability and reproducibility. The dataset produced in Phase 1 is fully prepared and will serve as the foundation for model training and evaluation in Phase 2.

1. Introduction

Stress is a universal human experience that affects individuals across academic, professional, and personal domains. When experienced chronically or at high intensity, stress is associated with a range of adverse health outcomes, including anxiety disorders, depression, impaired cognitive function, and cardiovascular disease. Early and reliable detection of stress levels can enable timely interventions and support, making automated stress monitoring a valuable area of applied research.

The proliferation of digital text data — spanning social media posts, forum discussions, journal entries, and survey responses — has created new opportunities for detecting psychological states through natural language. Text written under stress often exhibits distinctive linguistic features, including expressions of worry, fatigue, urgency, and emotional distress. Machine learning models trained on labeled text data can learn to identify these patterns and predict the associated stress level with reasonable accuracy.

This project is situated within the Medical Track of CSCI417 — Machine Intelligence and aims to develop a text classification system that assigns one of three stress level labels — Low, Medium, or High — to a given input text. The system operates purely on written language and does not require physiological or behavioral sensor data, making it broadly applicable in contexts such as mental health screening tools, student wellbeing monitoring, and workplace wellness applications.

The project is divided into two phases. Phase 1, documented in this report, focuses on the collection, processing, and documentation of the training dataset. Phase 2, to be completed subsequently, will involve feature extraction, model training, comparative evaluation, and the preparation of a final research paper.

2. Problem Statement

The problem addressed in this project is the automatic classification of written text into one of three predefined stress level categories. Given a text input representing a natural language response such as a journal entry, a social media post, a forum question, or a survey answer, the system should produce a stress level label indicating the degree of stress reflected in the text.

The task is formally defined as follows. The input to the system is a single text sample written in English. The output is one of three categorical labels: Low, Medium, or High, representing the inferred stress level of the author at the time of writing. This constitutes a supervised multi-class text classification problem, in which a model is trained on labeled examples and subsequently applied to unseen text samples.

Machine learning is a natural fit for this problem. The relationship between linguistic expression and psychological state is complex, context-dependent, and not easily captured by deterministic rules. A supervised learning approach enables the model to discover and generalize statistical patterns from labeled training data. Furthermore, having three balanced classes simplifies the evaluation process and allows the use of standard classification metrics to assess model performance.

3. Literature Review

3.1 Text Classification and Natural Language Processing

Text classification is one of the foundational tasks in natural language processing (NLP) and involves assigning predefined categories to text documents based on their content. Early approaches relied on hand-crafted features and statistical models such as Naive Bayes and Support Vector Machines, while more recent work has employed deep learning architectures and pre-trained language representations. The field has matured considerably, with robust methodologies available for a wide range of classification tasks including sentiment analysis, topic categorization, intent detection, and emotion recognition.

3.2 Stress and Mental Health Detection from Text

The detection of mental health states from written language has received increasing attention in the research community, particularly following the growth of social media platforms as rich sources of self-expressed psychological content. Studies have applied text classification techniques to datasets drawn from platforms such as Reddit, Twitter, and online forums to detect conditions including depression, anxiety, and general distress. A recurring finding in this body of work is that lexical features, writing style, and semantic content are informative predictors of the author's psychological state.

Stress, as a specific psychological construct, has been studied in text-based settings through datasets derived from diary entries, academic forums, and social media posts. Research has shown that linguistic markers such as negative affect words, cognitive load indicators, and expressions of temporal pressure correlate with self-reported stress levels. Feature engineering methods including Term Frequency–Inverse Document Frequency (TF-IDF), n-gram models, and word embeddings have all been demonstrated to capture aspects of stress-related language.

3.3 Machine Learning for Mental Health Classification

A variety of supervised learning algorithms have been applied to mental health text classification tasks. Classical models such as Logistic Regression, Naive Bayes, Decision Trees, Random Forests, and Support Vector Machines have demonstrated competitive performance on text classification benchmarks when combined with appropriate feature representations. More recent transformer-based models such as BERT have achieved state-of-the-art results on many NLP tasks; however, simpler models remain relevant for datasets of moderate size, where they often generalize well and are more interpretable.

The importance of balanced, well-curated datasets is consistently emphasized in the literature. Class imbalance, label ambiguity, and domain shift between training and deployment conditions are recognized challenges in mental health NLP. This project addresses these concerns through deliberate label selection, exclusion of clinically ambiguous categories, multi-source data collection, and dataset balancing.

4. Dataset Description

4.1 Overview

The final dataset used in this project is stored in the file `data/processed/stress_level_dataset_v2_balanced.csv`. It was constructed by combining data from two sources: the MentalDistress dataset and text collected from Stack Exchange via its public API. The final dataset is balanced, with equal representation across all three stress classes.

Class	Sample Count
Total samples	5,523
Low stress samples	1,841
Medium stress samples	1,841
High stress samples	1,841

4.2 Source Distribution

The contributions of each source to the final dataset are summarized in the table below. The MentalDistress dataset constitutes the majority of the data, while the Stack Exchange samples provide supplementary real-world text diversity.

Source	Contribution
MentalDistress dataset	4,419 samples
Stack Exchange API	1,104 samples
Total	5,523 samples

4.3 Dataset Columns

The final dataset contains fifteen columns designed to maximize traceability and metadata richness. These columns are listed below:

- `id` — unique identifier for each sample
- `text` — the cleaned text entry
- `stress_level` — the assigned class label: Low, Medium, or High
- `source_type` — indicates the origin category of the sample
- `source_name` — name of the specific source dataset or platform
- `source_url` — URL reference for the source, where applicable
- `source_record_id` — original record identifier from the source
- `original_label` — the original label before mapping
- `label_mapping_rule` — the rule used to assign the stress level
- `language` — language of the text entry
- `date_collected` — date on which the sample was collected
- `collector` — identifier of the team member who collected the sample
- `labeler` — identifier of the team member who assigned the label

- review_status — whether the sample has been reviewed and validated
- notes — any additional remarks about the sample

5. Data Collection

5.1 MentalDistress Dataset

The primary data source for this project is the MentalDistress dataset, a publicly available collection of text entries originally annotated with mental health-related labels. The dataset was accessed and downloaded in CSV format. Each entry in the dataset consists of a text field containing a short written response and a label field describing the associated mental or emotional state.

From the full set of labels available in the MentalDistress dataset, three were selected for inclusion in this project: Others, Frustrated, and Anxious. These three labels were chosen because they correspond most directly to the three target stress levels and do not carry clinical diagnostic connotations that would complicate the classification task. Two labels, Depressed and Suicidal, were deliberately excluded; the rationale for this exclusion is discussed in Section 6.

After label selection and mapping, an intermediate version of the dataset was saved as `data/interim/stress_level_dataset_v1.csv`, containing 4,500 balanced samples with 1,500 entries per class.

5.2 Stack Exchange API Collection

To increase the diversity and real-world representativeness of the dataset, additional text data was collected from Stack Exchange using its public API. Stack Exchange hosts numerous communities where users discuss a wide range of topics, including academic challenges, career difficulties, workplace pressures, and mental wellbeing. These communities provide a rich source of naturally occurring text that reflects everyday stress experiences.

The scraping process targeted posts related to themes including academic stress, deadlines, burnout, workplace pressure, time management, anxiety, and low confidence. Posts were retrieved, cleaned, and filtered to remove formatting artifacts, code blocks, and off-topic content. The cleaned Stack Exchange dataset was saved as `data/processed/scraped_stackexchange_stress_clean.csv`, containing 1,118 samples distributed across the three stress classes.

The inclusion of Stack Exchange data enhances the dataset by introducing text that differs stylistically from the MentalDistress entries. This diversity is expected to improve the generalization ability of models trained on the combined dataset.

6. Label Mapping

6.1 Mapping Scheme

The original labels in the MentalDistress dataset were mapped to stress level categories using the following scheme:

Original Label	Mapped Stress Level
Others	Low
Frustrated	Medium
Anxious	High

The label Others was interpreted as representing a neutral or mildly positive mental state, with no prominent indicators of distress, and was therefore assigned to the Low stress category. The label Frustrated was associated with a moderate level of emotional arousal and difficulty, corresponding to Medium stress. The label Anxious was associated with significant psychological distress, including worry and apprehension, and was assigned to the High stress category.

6.2 Excluded Labels

Two labels from the MentalDistress dataset — Depressed and Suicidal — were excluded from the project dataset. This decision was made deliberately and for principled reasons. Both labels refer to serious clinical mental health conditions that fall outside the scope of stress level classification. Including these labels in a stress classification dataset would introduce conceptual ambiguity, as depression and suicidal ideation are not simply extreme points on a stress continuum but rather distinct psychiatric conditions with their own diagnostic criteria.

Furthermore, treating these labels as proxies for stress could result in models that misclassify clinical distress as stress, which would be ethically problematic in any applied setting. The project maintains a clear boundary between stress level assessment and clinical mental health diagnosis.

For the Stack Exchange dataset, labels were assigned based on the thematic content of each post. Posts related to burnout, overwork, and high-pressure deadlines were labeled High; posts about mild frustration and moderate work or academic difficulty were labeled Medium; and posts related to general questions, mild concerns, or positive topics were labeled Low.

7. Data Preprocessing

The preprocessing pipeline was implemented using a series of Python scripts and applied to both data sources before merging. The following steps were performed.

7.1 Loading and Filtering

Both the MentalDistress dataset and the Stack Exchange collected data were loaded into pandas DataFrames. For the MentalDistress dataset, only rows with labels belonging to the three selected categories — Others, Frustrated, and Anxious — were retained. All other rows, including those labeled Depressed or Suicidal, were filtered out.

7.2 Text Cleaning

Each text entry was subjected to a cleaning procedure that included converting all text to lowercase for uniformity, removing leading and trailing whitespace, collapsing multiple consecutive spaces into single spaces, and removing formatting artifacts such as HTML tags, escape characters, and non-alphanumeric symbols not relevant to the semantic content. Rows containing empty or near-empty text after cleaning were removed.

7.3 Label Assignment and Standardization

Following the mapping scheme described in Section 6, each retained row was assigned a standardized stress level label of Low, Medium, or High. The original label was preserved in the original_label column for traceability. A label_mapping_rule column was also populated to document which mapping rule was applied to each row.

7.4 Metadata Enrichment

Each row was enriched with metadata columns including source_type, source_name, source_url, date_collected, collector, labeler, and review_status. This metadata structure ensures that the provenance of each entry can be traced and supports future auditing or dataset extension.

7.5 Merging

The processed MentalDistress dataset and the cleaned Stack Exchange dataset were merged into a single DataFrame. The combined dataset prior to balancing contained 5,594 samples, distributed as follows: 1,907 Low, 1,841 Medium, and 1,846 High. This combined file was saved as data/processed/stress_level_dataset_v2_combined.csv.

7.6 Deduplication

After merging, the combined dataset was checked for duplicate text entries. Any rows sharing identical text content were identified and removed to prevent data leakage and ensure that evaluation metrics during Phase 2 accurately reflect model generalization.

7.7 Balancing

To ensure equal representation across all three stress levels, the combined dataset was balanced by downsampling the two over-represented classes to match the size of the smallest class, which was Medium with 1,841 samples. The final balanced dataset contains exactly 1,841 samples per class, for a total of 5,523 samples. This balanced file was saved as `data/processed/stress_level_dataset_v2_balanced.csv`.

8. Dataset Creativity and Contribution

The dataset produced in this project was not simply adopted from a single existing source without modification. Rather, the team made a series of deliberate design decisions that constitute a meaningful original contribution.

First, the team curated the MentalDistress dataset by selecting only labels that are appropriate for stress level classification and excluding labels associated with clinical conditions. This required careful judgment about the conceptual boundaries between stress and mental illness.

Second, the team independently collected additional data from Stack Exchange using API-based scraping. This introduced a second modality of text — real-world forum discourse — that is stylistically different from the mental health survey responses in the MentalDistress dataset. Combining these sources increases the linguistic diversity of the training data and makes the resulting classifier more likely to generalize to varied real-world text inputs.

Third, the team designed a fifteen-column dataset schema that captures not only the text and label but also detailed metadata about each sample's origin, collection process, and labeling decisions. This level of documentation is uncommon in student projects and reflects a commitment to reproducibility and transparency.

Fourth, the dataset was balanced across all three classes, ensuring that no single stress level dominates the training data. A balanced dataset is an important prerequisite for training classifiers that perform reliably across all categories.

In summary, the final dataset represents an original, multi-source, balanced, and well-documented resource that goes beyond simply downloading and using a pre-existing dataset.

9. Challenges

9.1 Label Ambiguity and Boundary Definition

One of the primary challenges encountered during this phase was the difficulty of mapping mental health labels to stress level categories in a principled way. The distinction between Low, Medium, and High stress is inherently subjective, and the original labels in the MentalDistress dataset were not designed with a three-level stress taxonomy in mind. Determining that Others should represent Low stress and Frustrated should represent Medium stress required interpretive judgment that may not generalize universally.

9.2 Exclusion of Sensitive Labels

The decision to exclude Depressed and Suicidal labels required careful consideration. While including these entries might have increased the dataset size, doing so would have compromised the conceptual integrity of the task. Ensuring that the classification system does not inadvertently conflate stress with clinical mental health conditions was treated as a non-negotiable design constraint.

9.3 Text Noise and Inconsistency

Both data sources contained a degree of textual noise. The MentalDistress dataset included entries with inconsistent capitalization, informal abbreviations, and incomplete sentences. The Stack Exchange data contained additional challenges such as embedded code snippets, hyperlinks, markdown formatting, and HTML tags that required careful cleaning. Developing a robust preprocessing pipeline that handled both sources appropriately was a non-trivial task.

9.4 Cross-Source Consistency

Combining text from two structurally different sources introduced the risk of distributional mismatch. The language style, typical entry length, and vocabulary of the MentalDistress dataset differ from those of Stack Exchange posts. Ensuring that the merged dataset was coherent and that labels were consistently applied across both sources required careful review.

9.5 Dataset Balancing

Prior to balancing, the three stress classes in the combined dataset were slightly unequal in size. Achieving exact balance required downsampling the over-represented classes, which resulted in the removal of a small number of otherwise valid entries. An alternative strategy of oversampling minority classes was considered but was not applied at this stage, as downsampling was sufficient given the relatively small imbalance.

10. Future Work

Phase 2 of this project will focus on the implementation and evaluation of machine learning models for stress level classification using the dataset produced in Phase 1. The following activities are planned for Phase 2.

10.1 Advanced Text Preprocessing

Prior to model training, additional text preprocessing steps will be applied. These include tokenization, stop word removal, stemming or lemmatization, and the removal of any remaining noise. These steps will ensure that the text is in the most informative form before feature extraction.

10.2 Feature Extraction

The preprocessed text will be transformed into numerical feature representations suitable for machine learning algorithms. The primary feature extraction methods to be explored include Term Frequency–Inverse Document Frequency (TF-IDF) vectorization and Count Vectorization. Depending on time and resource availability, word embedding approaches may also be considered.

10.3 Model Training

At least five machine learning classification models will be trained and evaluated on the dataset. The candidate models currently under consideration include Logistic Regression, Naive Bayes, Support Vector Machine, Random Forest, Decision Tree, and K-Nearest Neighbors. Each model will be trained using an appropriate data split strategy, such as stratified train-test splitting or cross-validation, to ensure fair and reliable performance estimates.

10.4 Model Evaluation and Comparison

All trained models will be evaluated using standard classification metrics including accuracy, precision, recall, F1-score, and confusion matrices. Because the dataset is balanced, both macro-averaged and weighted metrics will be reported. The performance results of all models will be compiled into a comparative table, and the best-performing model will be identified based on overall F1-score and accuracy.

10.5 Final Research Paper

The complete methodology, experimental results, and conclusions from Phase 2 will be presented in a formal research paper. The paper will cover the full pipeline from dataset construction through model evaluation and will discuss the implications of the findings for text-based stress detection applications.

11. Conclusion

This report has documented the successful completion of Phase 1 of the Stress Level Classifier project. The primary deliverable of Phase 1 was a clean, balanced, and well-documented dataset suitable for training machine learning models to classify text into three stress level categories: Low, Medium, and High.

The final dataset, stored as `data/processed/stress_level_dataset_v2_balanced.csv`, contains 5,523 samples with equal class representation, consisting of 1,841 samples per stress level. The dataset was assembled from two complementary sources: the MentalDistress dataset, which contributed 4,419 samples, and Stack Exchange API data, which contributed 1,104 samples. A structured schema of fifteen columns ensures that each entry is fully traceable to its source and clearly documents the labeling decisions applied.

The label mapping scheme was designed with conceptual care, selecting only labels appropriate for stress level classification and explicitly excluding labels associated with clinical mental health conditions. The preprocessing pipeline addressed text cleaning, deduplication, and class balancing in a systematic manner. The dataset construction process reflects both technical rigor and thoughtful domain-aware decision-making.

Phase 1 has produced a dataset that is ready to support Phase 2 model training and evaluation. The foundation laid in this phase will directly enable the comparative study of multiple machine learning classifiers, the selection of the best-performing model, and the preparation of a final research paper documenting the complete project.

References

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